

Microeconomics **Problem Set 5**, Spring 2023

1. (Oligopoly) Suppose that two firms are Cournot competitors. Industry demand is given by $P = 200 - q_1 - q_2$, where q_1 is the output of Firm 1 and q_2 is the output of Firm 2. P is the price in dollars. Both Firm 1 and Firm 2 face constant marginal and average total costs equal to \$20.
- a. Solve for the Cournot price, quantity for each firm, and profits for each firm.

The derivation of Firm 1's reaction curve is:

$$\begin{aligned}MR_1 &= 200 - 2q_1 - q_2 \\MR_1 = MC_1 &\rightarrow 200 - 2q_1 - q_2 = 20 \leftrightarrow q_1 = 90 - 0.5q_2\end{aligned}$$

Since $MC_1 = MC_2$, Firm 2's reaction curve is symmetric to Firm 1's, or

$$q_2 = 90 - 0.5q_1$$

Then, substituting Firm 2's reaction function into Firm 1's,

$$\begin{aligned}q_1 &= 90 - 0.5(90 - 0.5q_1) \leftrightarrow q_1 = 45 + 0.25q_1 \leftrightarrow q_1 = 60 \\&\rightarrow q_2 = 90 - 0.5(60) = 60 \\&\rightarrow P = 200 - 60 - 60 = \$80 \\&\rightarrow \pi_1 = \pi_2 = (\$80 - 20) \times 60 = \$3600.\end{aligned}$$

- b. Firm 1 is considering investing in costly technology that will enable it to reduce its costs to \$14 per unit. What is the maximum amount Firm 1 should be willing to pay for this technology? Assume that Firm 1 can guarantee that Firm 2 will not be able to acquire it. (Hint: Think carefully about the additional benefit to Firm 1 if they have this technology vs if they do not)

We rework part (a) with $MC_1 = 14$, and figure out how much additional profit Firm 1 will make. This is the maximum amount they will be willing to pay.

$$MR_1 = MC_1 \rightarrow 200 - 2q_1 - q_2 = c \leftrightarrow q_1 = 93 - 0.5q_2$$

Firm 2's reaction function is still

$$q_2 = 90 - 0.5q_1$$

$$1. p = 200 - q_1 - q_2 \quad MC = AC = 20$$

a. 同時同質數量

$$\max \pi_1 = (200 - q_1 - q_2) \cdot q_1 - 20q_1$$

$$[q_1] = 200 - 2q_1 - q_2 - 20 = 0$$

$$\Rightarrow 180 - 2q_1 - q_2 = 0$$

$$\max \pi_2 = (200 - q_1 - q_2) \cdot q_2 - 20q_2$$

$$[q_2] = 180 - 2q_2 - q_1 = 0$$

$$\Rightarrow q_1^* = q_2^* = 60, \quad p^* = 200 - 120 = 80$$

$$\pi_1^* = \pi_2^* = (80 - 20) \cdot 60 = 3,600$$

$$\begin{array}{r} 180 \\ \underline{372} - 4q_1 - 2q_2 \\ \Rightarrow \end{array}$$

b. $\max \pi_1 = (200 - q_1 - q_2) \cdot q_1 - 14q_1$

$$[q_1] = 200 - 2q_1 - q_2 - 14 = 0 \Rightarrow 186 - 2q_1 - q_2 = 0$$

$$[q_2] = 180 - 2q_2 - q_1 = 0$$

$$180 - 2(186 - 2q_1) - q_1 = 0$$

$$180 - 372 + 4q_1 - q_1 = 0 \Rightarrow 3q_1 = 192 \Rightarrow q_1 = 64$$

$$q_2 = 58$$

$$p = (200 - 64 - 58) = 496$$

$$4096 - 3,600 = 496 \quad \#$$

$$c. \quad q_2 = 93 - 0.5q_1$$

$$\Rightarrow q_1 = 93 - 0.5(93 - 0.5q_1) \Rightarrow q_1 = 62 = q_2$$

$$p = 200 - 62 \times 2 = 76$$

$$\pi = (76 - 14) \times 62 = 3,844$$

$$3,844 - 3,600 = 244$$

Then, substituting Firm 2's reaction function into Firm 1's,

$$\begin{aligned} q_1 &= 93 - 0.5(90 - 0.5q_1) \leftrightarrow q_1 = 48 + 0.25q_1 \leftrightarrow q_1 = 64 \\ &\rightarrow q_2 = 90 - 0.5(64) = 58 \\ &\rightarrow P = 200 - 64 - 58 = \$78 \\ &\rightarrow \pi_1 = (\$78 - 14) \times 64 = \$4096 \end{aligned}$$

Since $\$4096 - 3600 = \496 , the firm would be willing to pay \$496 to invest in the technology to lower cost.

- c. How does your answer to (b) change if Firm 1 knows that Firm 2 can and will invest in the new technology at the same time Firm 1 does?

Now, Firm 2 has the same reaction function as Firm 1 in part (b), so

$$q_2 = 93 - 0.5q_1$$

Then, substituting Firm 2's reaction function into Firm 1's,

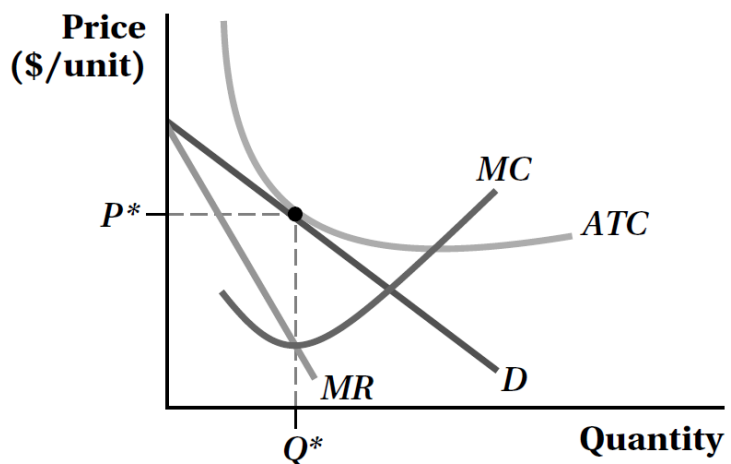
$$\begin{aligned} q_1 &= 93 - 0.5(93 - 0.5q_1) \leftrightarrow q_1 = 46.5 + 0.25q_1 \leftrightarrow q_1 = 62 \\ &\rightarrow q_2 = 93 - 0.5(62) = 62 \\ &\rightarrow P = 200 - 62 - 62 = \$76 \\ &\rightarrow \pi_1 = (\$76 - 14) \times 62 = \$3844 \end{aligned}$$

Since $\$3844 - 3600 = \244 , Firm 1 will only be willing to pay \$244 for the new technology.

2. (Oligopoly) Consider a monopolistically competitive industry. A graph of demand and cost conditions for a typical firm is depicted in the diagram below.

- a. Give an example of a monopolistically competitive industry and explain how your example fits the model's assumptions.

Brands of cereal are one such example. Consumers do not view different cereals as



2.

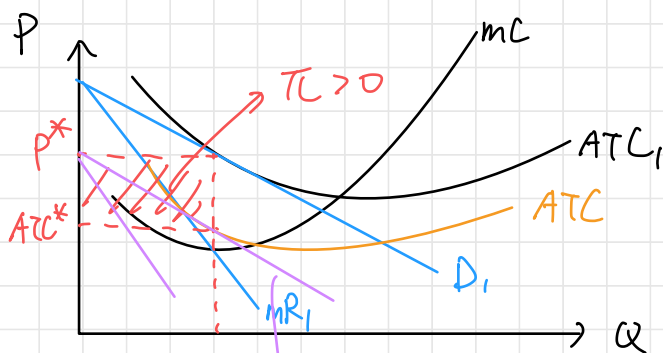
a. 各大通訊業者

b. firms earn zero economic profit b/c $P^* = ATC$

$$\text{at } Q^* \Rightarrow \pi^* = (P - ATC) \times Q^* = 0$$

c. There will not expect to have entry or exit occur.

d. ATC shifts down, b/c FC reduces. But P^* and Q^* remain unchanged, b/c MR, MC do not change
at when at Q^* $P > ATC \Rightarrow \pi^* > 0$



expect entry / MR shift in and become flatter

perfect substitutes, so an individual producer is likely to face a downward sloping demand curve. There are likely low or no barriers to entry or exit, however, so competition drives down economic profits in the long-run.

- b. What is the economic profit earned by the firm represented in the graph?

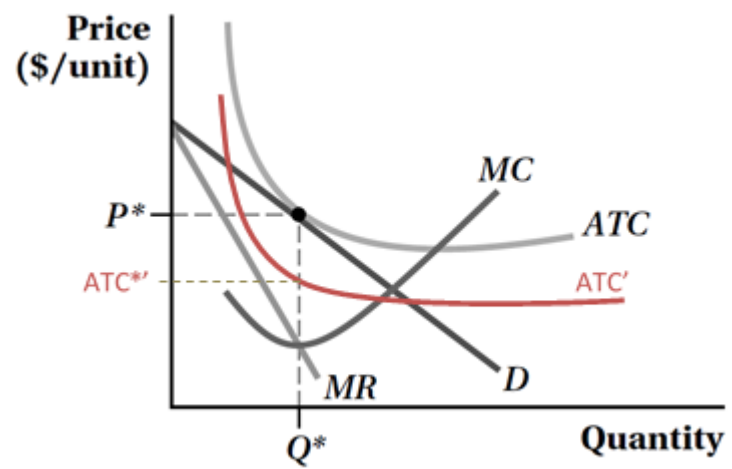
This firm is earning zero economic profit because P^* is equal to ATC at Q^* , which implies $\pi^* = (P^* - ATC^*) \times Q^* = 0 \times Q^* = 0$.

- c. Do you expect any entry into or exit from this industry to occur? Explain.

Since the representative firm is earning zero economic profit, we do not expect any entry or exit to occur.

- d. Suppose that the government reduces annual licensing fees, causing the fixed cost of the typical firm to fall. Make appropriate shifts of all curves that might be affected. What happens to profit?

ATC shifts down because FC are lower. P^* and Q^* do not change because MR and MC have not changed, but now at Q^* , $ATC < P$, which means profit has increased.



- e. Do you expect the fall in fixed costs to cause entry into or exit from this industry? Explain. In words, which curves will this change and how?

Because economic profit is now positive, we expect entry into the industry and for demand and MR to shift in and become flatter, until the new demand curve is tangent to the new ATC curve.

3. (Uncertainty) You purchase a new smartphone for \$650. Suppose that during one year, there is a 0.2 probability that you will drop the phone, a 0.2 probability you “send it for a swim”

(drop it in water), and a 0.6 probability you avoid all accidents. If you drop the phone, you will crack the screen, which must be repaired for \$150. If you drop it in water, it will be completely broken and you will have to replace it for the full cost. Assume that forgoing the phone, or using one with a cracked screen, is not an option.

- a. What is your expected loss from one year of smartphone ownership?

$$E(\text{Loss}) = 0.2 \times \$150 + 0.2 \times \$650 + 0.6 \times 0 = \$30 + \$130 = \$160$$

- b. The sales associate at the store offers to sell you an extended warranty that costs \$130. If you damage or destroy your phone accidentally, the store will repair your phone or send you a new one. Assuming you are risk neutral, should you buy the warranty?

Yes, if you are risk neutral, you should buy the warranty because your expected loss is greater than the cost of the warranty.

- c. Suppose the warranty costs \$130, but in the event of either a cracked screen or broken phone, you must pay a \$99 “service fee” before the store repairs your phone or sends you a new one. Assuming you are risk neutral, should you buy the warranty now? (Hint: Start by re-doing part (a) for the expected loss you will suffer if you have the warranty)

$$\begin{aligned} E(\text{Loss}_{\text{warranty}}) &= 0.2 \times \$99 + 0.2 \times \$99 + 0.6 \times 0 \\ &= \$19.80 + \$19.80 = \$39.60 \end{aligned}$$

The warranty lower's your expected loss by \$120.40, but costs you \$130 to do so, so you should not buy the warranty.

- d. Now let's suppose you are risk neutral, but especially accident prone. Now there is a 0.3 probability you will crack the screen, 0.3 probability you will drown your phone, and 0.4 probability you will avoid all accidents. Should you buy the warranty as described in part (c)?

$$\begin{aligned} E(\text{Loss}_{\text{nowarranty}}) &= 0.3 \times \$150 + 0.3 \times \$650 + 0.4 \times 0 \\ &= \$45 + \$195 + 0 = \$240 \end{aligned}$$

$$E(\text{Loss}_{\text{warranty}}) = 0.3 \times \$99 + 0.3 \times \$99 + 0.4 \times 0 = \$59.40$$

Now, the warranty lower's your expected loss by \$180.60, but costs you only \$130, so you should buy it.

- e. Suppose you are not accident-prone. Can you think of any other situation where it might make sense for you to buy the extended warranty?

$$\begin{array}{l} \lambda \text{ a.} \\ \begin{array}{l} 0.2 \rightarrow -150 \\ 0.2 \rightarrow -650 \\ 0.6 \rightarrow 0 \end{array} \end{array}$$

$$\begin{aligned} \text{expect loss} &= 0.2 \cdot 150 + 0.2 \cdot 650 + 0.6 \cdot 0 \\ &= 160 \end{aligned}$$

b. Yes, because risk neutral, the expected loss is greater than the warranty cost. Buy!

$$\begin{aligned} \text{c. expect loss (warranty)} &= 0.2 \cdot 99 + 0.2 \cdot 99 + 0.6 \cdot 0 \\ &= 39.60 \end{aligned}$$

$$\begin{aligned} \text{the cost of warranty total expect cost} &= 130 + 39.60 \\ &= 169.60 \end{aligned}$$

$\Rightarrow$ greater than the expected loss. Not Buy!

$$\begin{aligned} \text{d. expect loss} &= 0.3 \cdot 150 + 0.3 \cdot 650 + 0.4 \cdot 0 \\ &= 240 \end{aligned}$$

$$\begin{aligned} \text{warranty cost} &= 130 + 0.3 \cdot 99 + 0.3 \cdot 99 \\ &= 189.4 \end{aligned}$$

$\Rightarrow$ Buy the warranty!

e.

If you are risk-averse enough, your expected utility from buying the warranty might be higher than your expected utility from not buying the warranty, even though your expected income/wealth is lower by purchasing the warranty.

4. (Uncertainty) Danielle is a farmer, with a utility function of $u(m) = \sqrt{m}$, where u is Danielle's utility and m is the realization of her income. If the weather is good, she will earn \$100,000. If there is a hailstorm, she will earn only \$50,000. The probability of a hailstorm in any given year is 30%.

- a. What is Danielle's expected income if she is uninsured? Her expected utility?

$$\text{Expected income: } E(M) = 0.7 \times \$100,000 + 0.3 \times \$50,000 = \$70,000 + \$15,000 = \$85,000$$

$$\begin{aligned} \text{Expected utility: } E(u(M)) &= 0.7 \times \sqrt{\$100,000} + 0.3 \times \sqrt{\$50,000} \\ &= 0.7 \times 316.2278 + 0.3 \times 223.6068 \approx 288.441 \end{aligned}$$

- b. Suppose a crop insurer makes the following offer to Danielle: Danielle pays an annual premium of \$15,000 no matter the weather, and in years when there is a hailstorm, the insurer pays Danielle \$48,000. What is Danielle's expected income? Her expected utility?

$$\text{Income (no hailstorm): } m_1 = \$100,000 - \$15,000 = \$85,000$$

$$\text{Income (hailstorm): } m_2 = \$50,000 - \$15,000 + \$48,000 = \$83,000$$

$$\text{So expected income is } E(M) = 0.7 \times \$85,000 + 0.3 \times \$83,000 = \$84,400$$

$$\begin{aligned} \text{Expected utility: } E(u(M)) &= 0.7 \times \sqrt{\$85,000} + 0.3 \times \sqrt{\$83,000} \\ &= 0.7 \times 291.5476 + 0.3 \times 288.0972 \approx 290.512 \end{aligned}$$

- c. Comment on the following statement referring to your answers to parts (a) and (b): "The insurance agreement in (b) reduces Danielle's expected income. Therefore, it must make her worse off."

Although the insurance agreement reduces her expected income, it actually raises her expected utility by decreasing her risk, so she is better off. The reason is that she is risk-averse, so she is willing to trade some expected utility for reduced risk.

- d. What is Danielle's risk premium? That is, if she has no insurance, how much of her expected income would she be willing to give up to eliminate her risk?

We first calculate Danielle's certainty equivalent, which we get by finding the square of her expected utility from part a

$$u(m_{ce}) = \sqrt{m_{ce}} = 288.441 \rightarrow m_{ce} = 288.441^2 \approx \$83,198.48$$

This implies the risk premium is $\$85,000 - \$83,198.48 \approx \$1,801.52$

4. a. expect income = $100,000 \times 70\% + 50,000 \times 30\%$

$$= 70,000 + 15,000 = 85,000$$

$$\text{expect utility} = 0.7 \times \sqrt{100,000} + 0.3 \times \sqrt{50,000}$$

$$= 70\sqrt{10} + 30\sqrt{5}$$

b. expect income : $100,000 - 15,000 = 85,000$

$$85,000 \times 0.7 + 83,000 \times 0.3 = 84,400$$

$$\text{expect utility} = 0.7 \times \sqrt{100,000} + 0.3 \times \sqrt{83,000}$$

$$50,000 - 15,000 + 48,000 = 83,000$$

d. risk premium / risk averse

$$u(m) = \sqrt{m} = 70\sqrt{10} + 30\sqrt{5}$$

$$\Rightarrow m = (70\sqrt{10} + 30\sqrt{5})^2 = 83,198.48$$

$$85,000 - 83,198.48 = 1,801.52$$

預期的效用應有多少的 expected income

再用沒有保險的 expected income - 效用應有的 expected income